

SIMPLER GRAPH CONDITIONS FOR EMBEDDING TETRAHEDRAL MESHES

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ABSTRACT. Three-dimensional Tutte-style embedding methods turn graph conditions into guarantees for tetrahedral meshes. Alexa’s theorem excludes both K_6 and $K_{3,3,1}$ as graph minors and asks whether the second exclusion is necessary. We prove that it is redundant for a precisely defined class of topological-ball meshes. Let T be a finite simplicial complex whose realization is a closed topological 3-ball, and assume that every triangular face whose vertices all lie on the boundary is itself a boundary face. If the 1-skeleton G has no K_6 minor, then G is linklessly embeddable and hence has no $K_{3,3,1}$ minor.

The proof combines generic 4-rigidity and Jørgensen’s extremal classification with a four-clique separator bound obtained from relative Alexander–Lefschetz duality. The Holst–Lovász–Schrijver clique-sum criterion then preserves linklessness throughout the resulting MP_1 -cockade decomposition. A finite checker and a scoped Lean companion audit the local separator calculation and logical interfaces; the imported rigidity, extremal-minor, and linkless-embedding theorems remain external. The result assumes an actual topological ball and does not extend merely from simple connectivity.

1. INTRODUCTION

Alexa proved a three-dimensional analogue of Tutte’s barycentric embedding theorem for tetrahedral meshes [2]. Besides convexity and connectivity assumptions, his theorem requires that the mesh graph contain neither K_6 nor $K_{3,3,1}$ as a minor. The paper reports that no mesh with a $K_{3,3,1}$ minor but no K_6 minor could be made to demonstrate the necessity of the second exclusion, and asks whether that clause is needed [2, Section 5].

There is a source-language point that matters here. The displayed theorem says that the boundary is simply connected, while the setup separately says that the tetrahedral mesh is assumed to be a topological ball [2, Theorem 1.1 and Section 2]. We do not identify these formulations. Our structural theorem assumes an actual closed topological 3-ball. Consequently, its Tutte-embedding corollary applies to the topological-ball setting and makes

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no claim for a broader object satisfying only a weaker simply-connectedness condition.

Let ∂T denote the boundary subcomplex. The condition inherited from Alexa is

$$(BT) \quad \sigma \in T, \quad \dim \sigma = 2, \quad V(\sigma) \subseteq V(\partial T) \implies \sigma \in \partial T.$$

It concerns triangular simplices, not arbitrary three-cycles in the graph.

Theorem 1.1 (Structural theorem). *Let T be a finite simplicial complex whose realization is a closed topological 3-ball, and let $G = T^{(1)}$. If T satisfies (BT) and G has no K_6 minor, then G is linklessly embeddable. In particular, G has no $K_{3,3,1}$ minor.*

Theorem 1.1 does not use the convex-combination realization or four-connectivity. They return only in the application.

Corollary 1.2 (Tutte-embedding consequence). *Let T be a tetrahedral triangulation of a closed topological 3-ball satisfying (BT). Assume that $G = T^{(1)}$ is 4-connected and has no K_6 minor, that the boundary triangles realize a strictly convex polyhedron in $\mathbb{R}^3$, and that every interior vertex is a strict convex combination of its graph neighbors. Then the realized tetrahedral mesh is embedded.*

The proof combines four ingredients. First, (BT) makes the interior vertices connected and their closed stars cover the graph. Gluck’s theorem, coning, and rigidity gluing make the skeleton generically 4-rigid [3, 4]. Second, the lower edge count from rigidity meets Mader’s K_6 -minor bound [8]; Jørgensen’s equality classification then gives an MP_1 -cockade [6, 13]. Third, duality and a four-vertex relative calculation show that deletion of any four-clique leaves at most two components. Finally, the clique-sum criterion of Holst, Lovász, and Schrijver, in the precise form recorded by Naimi, Pavelescu, and Pavelescu, rules out intrinsic linking at every K_4 -sum [19, 11].

The finite certificate checks, but does not replace, the hand proof of the four-vertex relative calculation. Lean checks the finite Boolean cycle-count certificate and the logical composition of explicitly assumed mathematical inputs. It does not formalize homology, rigidity, duality, graph minors, spatial embeddings, the Holst–Lovász–Schrijver theorem, or Alexa’s theorem. Thus the proof is software-independent and the formal scope is explicit.

2. DEFINITIONS AND EXTERNAL INPUTS

All graphs are finite and simple. A graph is *intrinsically linked* (IL) if every tame embedding in $\mathbb{R}^3$ contains a nonsplit link formed by two vertex-disjoint cycles; it is *linklessly embeddable*, or nIL, otherwise. Linkless embeddability is minor closed. By the Robertson–Seymour–Thomas theorem, the excluded minors are the seven graphs in the Petersen family [16]. In particular, $K_{3,3,1}$ is IL.

For a simplex of T , *interior* means “not contained in the boundary sub-complex ∂T ”; thus an interior edge may have two boundary vertices. Because $|T|$ is a 3-manifold with boundary, T is pure: every simplex is a face of a tetrahedron. Indeed, a maximal simplex of dimension less than three would give a relative-interior point with no three-dimensional manifold neighborhood. We use the standard PL facts about manifold links and general position recorded by Rourke and Sanderson [17, Chs. 2 and 5].

A graph is *generically d -rigid* if its generic bar-and-joint realization in $\mathbb{R}^d$ is infinitesimally rigid [4, Chs. 2 and 5]. We use the following standard facts:

- (R1) the graph of a triangulated 2-sphere is generically 3-rigid by Gluck’s theorem [3];
- (R2) coning a generically d -rigid graph produces a generically $(d+1)$ -rigid graph [20, 21];
- (R3) two generically d -rigid graphs whose intersection contains at least d vertices have generically d -rigid union.

For (R3), take the common vertices in generic position. An infinitesimal motion restricts to an infinitesimal Euclidean motion on each rigid piece. Choose d affinely independent common points $p_1, \dots, p_d$. The difference of the two motions has the form $x \mapsto a + \Omega x$, where $\Omega^T = -\Omega$, and vanishes at every p_i . Hence $\Omega(p_i - p_1) = 0$ for $i = 2, \dots, d$, so $\ker \Omega$ contains a $(d-1)$ -dimensional subspace and $\text{rank } \Omega \leq 1$. A real skew-symmetric matrix has even rank; therefore $\Omega = 0$, and then $a = 0$. The two infinitesimal Euclidean motions agree. This also explains why d , rather than $d+1$, common generic vertices suffice. For $n \geq 5$, generic 4-rigidity forces at least $4n - \binom{5}{2} = 4n - 10$ edges. Nevo’s complementary rigidity-matroid theorem says that every K_6 -minor-free graph is generically 4-stress-free [12, Thm. 1.3]. Thus the rigidity and minor hypotheses also say that the skeleton is a basis of the generic 4-rigidity matroid. We retain the Mader route below because it exposes the exact extremal inequality used by Jørgensen’s classification.

Mader’s extremal theorem gives the opposing inequality.

Theorem 2.1 (Mader [8]). *If a simple graph on $n \geq 5$ vertices has no K_6 minor, then it has at most $4n - 10$ edges.*

The equality case is described by MP_1 -cockades.

Theorem 2.2 (Jørgensen [6]). *If a graph G has $n \geq 5$ vertices, $4n - 10$ edges, and no K_6 minor, then G is an MP_1 -cockade.*

Jørgensen proves the corresponding formula for $5 \leq p \leq 7$; only $p = 6$ is used here. An accessible restatement is [13, Thm. 5].¹

Definition 2.3. An MP_1 -cockade is generated recursively from K_5 and graphs $H * K_1$, where H is a 4-connected maximal planar graph, by clique sums over K_4 using disjoint copies before identification, as in [13, Def. 6].

¹The displayed binomial in that restatement is misprinted as $\binom{p}{2}$; its preceding Mader theorem and its immediate $p = 6$ application use the correct $\binom{p-1}{2}$, giving $4n - 10$.

The two atoms are apex graphs and hence nIL. For the induction we use the following exact consequence of the Colin de Verdière theory.

Theorem 2.4 (Holst–Lovász–Schrijver). *Let G be the clique sum over a clique S of two nIL graphs. Then G is IL if and only if two or three components of $G - S$ can be contracted to vertices so that, together with S , the resulting graph is K_7 minus the three edges of a triangle.*²

This is the graph form recorded in [11, Thm. 7] of the clique-sum theorem [19, Thm. 2.10]. The passage from the μ -statement to linkless embeddability uses $\mu(G) \leq 4 \iff G$ is nIL [7, 15].

Corollary 2.5 (Four-clique specialization). *Let G be a K_4 -sum of two nIL graphs over S . If $G - S$ has at most two components, then G is nIL.*

Proof. Each selected component is contracted to one vertex and all other components are discarded. With $|S| = 4$, the contracted graph in Theorem 2.4 therefore has at most

$$4 + \text{comp}(G - S) \leq 6$$

vertices, and a graph on six vertices has no minor on seven vertices. $\square$

3. RIGIDITY FORCES THE EXTREMAL EQUALITY

We first show that the closed stars of interior vertices cover a connected rigid skeleton.

Lemma 3.1 (Low-dimensional manifold links). *In a finite simplicial triangulation of a topological 3-manifold with boundary, the link of an interior vertex is a triangulated 2-sphere and the link of an interior edge is a triangulated circle.*

Proof. The dimension-three combinatorial-triangulation theorem says that every simplicial triangulation of a topological 3-manifold is combinatorial; the boundary case follows by doubling along the boundary [9, 10]. This is a dimension-specific theorem, not merely an inference from existence of some PL structure. In a combinatorial manifold the link of a simplex outside the boundary is a PL sphere [17, Ch. 2]. Thus the link of an interior k -simplex is a PL sphere of dimension $2 - k$. Taking $k = 0$ and $k = 1$ gives the two assertions. In particular, an interior edge link is a simplicial circle and therefore has at least three vertices. $\square$

Lemma 3.2. *If T satisfies (BT) and is not a single tetrahedron, then every tetrahedron of T contains an interior vertex.*

Proof. Suppose that a tetrahedron τ has four boundary vertices. Each of its four triangular facets has three boundary vertices, so (BT) places all four facets in ∂T . No other tetrahedron can share a facet with τ . The facet-dual

²The alternative “two or three” covers different clique sizes: for a five-clique, two contracted components already produce seven vertices.

graph is connected: choose interior points in any two tetrahedra and join them by a piecewise-linear path in the connected manifold interior. After a general-position perturbation, the path avoids the simplicial 1-skeleton and meets the 2-skeleton transversely in finitely many interior points of triangular faces. Reading the tetrahedra crossed by the path gives a facet-dual walk using only shared interior triangles; this is the standard PL general-position move in [17, Ch. 5]. A boundary triangle belongs to one tetrahedron and an interior triangle to two, so the crossings give a facet-dual walk. Hence τ is the only tetrahedron, contrary to the hypothesis. $\square$

Lemma 3.3. *Under the hypotheses of Lemma 3.2, the subgraph induced by the interior vertices is connected.*

Proof. Two tetrahedra adjacent in the facet-dual graph share an interior triangular face. By (BT), that face cannot have three boundary vertices; it contains an interior vertex. By Lemma 3.2, the interior vertices of every tetrahedron form a nonempty clique. Along a path in the facet-dual graph, consecutive such cliques overlap. Their union is therefore connected and contains every interior vertex. $\square$

Lemma 3.4. *For every interior vertex v , the graph of the closed star $\text{st}_T(v)$ is generically 4-rigid.*

Proof. By Lemma 3.1, the link $\text{lk}_T(v)$ is a triangulated 2-sphere. Its graph is generically 3-rigid by Gluck's theorem [3]. Since $\text{st}_T(v) = v * \text{lk}_T(v)$, Whiteley's cone theorem [20, 21] gives generic 4-rigidity. $\square$

Proposition 3.5. *If T is not a single tetrahedron and satisfies (BT), then $G = T^{(1)}$ is generically 4-rigid.*

Proof. Choose a spanning tree of the connected interior-vertex graph from Lemma 3.3. Adjacent interior vertices u, v are joined by an edge uv , which cannot lie in ∂T ; hence $\text{lk}_T(uv)$ is a triangulated circle and has at least three vertices by Lemma 3.1. Hence

$$V(\text{st}_T(u)) \cap V(\text{st}_T(v)) \supseteq \{u, v\} \cup V(\text{lk}_T(uv)),$$

so the two star graphs share at least five vertices. Lemma 3.4 and (R3) show inductively along the spanning tree that their union is generically 4-rigid.

Purity gives that every edge of G belongs to a tetrahedron. By Lemma 3.2, that tetrahedron has an interior vertex, and the edge belongs to its closed star. Thus the union of the interior stars is exactly G . $\square$

Corollary 3.6. *If T is not a single tetrahedron, satisfies (BT), and $K_6 \not\leq G$, then*

$$|E(G)| = 4|V(G)| - 10,$$

and G is an MP_1 -cockade.

Proof. A nontrivial T contains at least two tetrahedra, which share at most a triangle, and hence has at least five vertices. Proposition 3.5 gives the lower

bound $4|V(G)| - 10$, while Theorem 2.1 gives the upper bound. Equality follows, and Theorem 2.2 applies. $\square$

4. FOUR-CLIQUE TOPOLOGY

The second new ingredient is a component bound after deleting an arbitrary four-clique. A graph clique need not span a tetrahedron; the argument allows all possible induced subcomplexes on its four vertices. By Lemma 3.1, T is combinatorial, so $|T|$ is a PL closed 3-ball and the induced complexes used below are subpolyhedra.

Proposition 4.1 (Deletion retraction). *Let $S \subseteq V(T)$, put $K = T[S]$, and write $T - S = T[V(T) \setminus S]$. Then $|T| \setminus |K|$ strongly deformation retracts onto $|T - S|$.*

Proof. For $x \in |T|$, let $\lambda_v(x)$ be its global piecewise-linear barycentric coordinate at v , and set $a(x) = \sum_{s \in S} \lambda_s(x)$. Because K is induced, $x \notin |K|$ exactly when $a(x) < 1$. For $0 \leq t \leq 1$, define

$$H_t(x) = \sum_{s \in S} \frac{(1-t)\lambda_s(x)}{1-ta(x)} s + \sum_{v \notin S} \frac{\lambda_v(x)}{1-ta(x)} v.$$

The denominator is positive. The coefficients are nonnegative, sum to one, and have support in the simplex containing x , so the homotopy remains in $|T| \setminus |K|$. At $t = 1$, all coordinates indexed by S vanish; points of $|T - S|$ are fixed throughout. $\square$

Proposition 4.2 (Relative ball duality). *Let B be a PL closed 3-ball and let $A \subset B$ be a nonempty compact subpolyhedron. Then*

$$(1) \quad \tilde{H}_0(B \setminus A; \mathbb{F}_2) \cong H^2(A, A \cap \partial B; \mathbb{F}_2).$$

Proof. If $B \setminus A = \emptyset$, then $A = B$, and both groups in (1) vanish. Hence assume $B \setminus A \neq \emptyset$. This is the relative ball form of Alexander–Lefschetz duality. We give the reduction to the standard sphere theorem [5, Thm. 3.44]; see also [18, Ch. 6, §9]. Choose a PL identification of B with one closed hemisphere of S^3 and put $E = \overline{S^3 \setminus B}$, so E is manifestly the other closed PL ball and $E \cap B = \partial B$. Collapsing E identifies $(A \cup E)/E$ with $A/(A \cap \partial B)$, and relative cohomology gives

$$H^2(A \cup E, E; \mathbb{F}_2) \cong H^2(A, A \cap \partial B; \mathbb{F}_2).$$

Because E is a nonempty ball, the long exact sequence of the pair gives $H^2(A \cup E, E; \mathbb{F}_2) \cong \tilde{H}^2(A \cup E; \mathbb{F}_2)$. The compact polyhedron $A \cup E$ is locally contractible, so Alexander duality in S^3 now gives

$$\tilde{H}^2(A \cup E; \mathbb{F}_2) \cong \tilde{H}_0(S^3 \setminus (A \cup E); \mathbb{F}_2) = \tilde{H}_0(\overset{\circ}{B} \setminus A; \mathbb{F}_2).$$

Simplicial, singular, and Čech cohomology agree for these polyhedra.

It remains to compare the two complements. Both are locally path connected: after a common subdivision, complements of compact subpolyhedra in a PL manifold are locally contractible. Thus their connected components,

path components, and the basis elements recorded by singular H_0 agree. The inclusion $\mathring{B} \setminus A \hookrightarrow B \setminus A$ is bijective on components. For surjectivity, if $x \in (B \setminus A) \cap \partial B$, the open set $B \setminus A$ contains a relative half-ball neighborhood of x , hence an interior point joined to x inside that neighborhood. For injectivity, let a path $\gamma \subset B \setminus A$ join two interior points. Its compact image has positive distance from the compact polyhedron A . Pushing the portion of γ in a sufficiently thin boundary collar inward by less than this distance, while fixing the two endpoints, produces a path in $\mathring{B} \setminus A$. The PL collar theorem supplies the collar [17, Ch. 3]. This proves (1). $\square$

Lemma 4.3 (Four-vertex relative bound). *Let S be a four-clique in G , put $K = T[S]$, and put $K_\partial = K \cap \partial T$. If T satisfies (BT), then*

$$\dim_{\mathbb{F}_2} H_2(K, K_\partial; \mathbb{F}_2) \leq 1.$$

Proof. The complex K contains the full 1-skeleton of the abstract tetrahedron on S . It contains an arbitrary subset of the four triangular faces, and it contains the 3-simplex only if all four faces are present. Let $b = |S \cap V(\partial T)|$.

If $b \leq 2$, then $H_1(K_\partial; \mathbb{F}_2) = 0$. The long exact sequence of the pair makes $H_2(K) \rightarrow H_2(K, K_\partial)$ surjective. A subcomplex of an abstract tetrahedron has second homology of dimension at most one: the only possible nonzero 2-cycle is the sum of all four triangular faces, and the tetrahedron, if present, kills it.

Suppose $b = 3$. We have $\dim H_1(K_\partial) \leq 1$. If this group vanishes, the preceding argument applies. Otherwise K_∂ contains the three boundary edges as a cycle but not the triangular face. By (BT), that face is absent from K . A nonzero 2-cycle would require all four faces, so $H_2(K) = 0$. Exactness injects $H_2(K, K_\partial)$ into $H_1(K_\partial)$.

Finally, if $b = 4$, every triangular face of K has three boundary vertices and belongs to K_∂ by (BT). Thus $C_2(K, K_\partial; \mathbb{F}_2) = 0$, and the relative group vanishes. $\square$

Theorem 4.4 (Four-clique separator bound). *If S is a four-clique of G and $G - S \neq \emptyset$, then*

$$\text{comp}(G - S) \leq 2.$$

Proof. Let $K = T[S]$. Proposition 4.1 identifies the components of $|T| \setminus |K|$ with those of $|T - S|$, and the latter are the components of its 1-skeleton $G - S$. By (1), the universal coefficient theorem over $\mathbb{F}_2$, and Lemma 4.3,

$$\begin{aligned} \text{comp}(G - S) - 1 &= \dim_{\mathbb{F}_2} \tilde{H}_0(|T| \setminus |K|; \mathbb{F}_2) \\ &= \dim_{\mathbb{F}_2} H^2(K, K_\partial; \mathbb{F}_2) \\ &= \dim_{\mathbb{F}_2} H_2(K, K_\partial; \mathbb{F}_2) \leq 1. \end{aligned}$$

$\square$

5. SAFE CLIQUE-SUM INDUCTION

Lemma 5.1. *Every apex graph is linklessly embeddable.*

Proof. Embed the planar graph obtained by deleting an apex a on a 2-sphere, place a in the enclosed ball, and join it to its neighbors by pairwise interior-disjoint radial arcs. Of two vertex-disjoint cycles, at most one contains a . Two cycles on the sphere are unlinked. If one contains a , deleting a leaves a path on the sphere disjoint from the other cycle; its cone to a is a spanning disk disjoint from that cycle. $\square$

The separator property in Theorem 4.4 belongs to the final graph G , while the HLS theorem is applied at internal nodes of a fixed cockade decomposition. The following elementary observation bridges that gap.

Lemma 5.2 (Attachments do not merge old components). *Let X and Y be graphs with $X \cap Y = R$, where R is a clique, and let S be a clique of X . Two vertices of $X - S$ are connected in $(X \cup Y) - S$ if and only if they are connected in $X - S$.*

Proof. Only the forward direction requires proof. Given a path in $(X \cup Y) - S$ with endpoints in $X - S$, replace every maximal excursion through $Y - X$ by an edge between its entry and exit vertices. If entry and exit coincide, delete the closed excursion instead. Otherwise those two vertices lie in $R - S$, and the required edge lies in $X - S$ because $R - S$ is a clique. If $R \subseteq S$, no such excursion can occur. The resulting walk lies in $X - S$ and contains a path with the same endpoints. $\square$

Proposition 5.3. *Let G be an MP_1 -cockade such that deletion of every four-clique leaves at most two components whenever the deletion is nonempty. Then G is linklessly embeddable.*

Proof. Fix an MP_1 -construction tree. Induct over its nodes. A leaf is K_5 or $H * K_1$ with H maximal planar; both are apex and nIL by Lemma 5.1.

Let an internal node C be the K_4 -sum of already nIL child graphs over S . Each child has at least five vertices (the atoms do, and sums only grow), so C has at least six vertices and $C - S \neq \emptyset$; a fortiori $G - S \neq \emptyset$. Let $C = C_0 \subset C_1 \subset \dots \subset C_k = G$ be the unions encountered on the path from this node to the root. Then $C_{i+1} = C_i \cup Y_{i+1}$ and $C_i \cap Y_{i+1} = R_{i+1} \cong K_4$, by the disjoint-copy clause of Definition 2.3. Apply Lemma 5.2 at each step with $X = C_i$, $Y = Y_{i+1}$, $R = R_{i+1}$, and with the original sum clique S , which remains a clique of C_i . Thus distinct components of $C - S$ inject into distinct components of $G - S$. Hence $\text{comp}(C - S) \leq \text{comp}(G - S) \leq 2$. Corollary 2.5 therefore makes C nIL. The induction reaches G . $\square$

6. PROOFS OF THE MAIN RESULTS

Proof of Theorem 1.1. If T is one tetrahedron, then $G = K_4$ is nIL. Otherwise, Corollary 3.6 makes G an MP_1 -cockade. Theorem 4.4 supplies the two-component condition for every four-clique, so Proposition 5.3 makes G linklessly embeddable. Since linkless embeddability is minor closed and $K_{3,3,1}$ is in the Petersen family, $K_{3,3,1} \not\preceq G$. $\square$

Proof of Corollary 1.2. Theorem 1.1 supplies the missing $K_{3,3,1}$ -minor exclusion. Because a closed topological 3-ball has boundary S^2 , both its boundary and its interior are simply connected. Together with (BT), four-connectivity, the two minor exclusions, strict boundary convexity, and the strict convex-combination condition, this matches the explicit topological-ball setting and the hypotheses stated in the corollary. No claim is made for a mere pseudomanifold satisfying only the displayed incidence conditions in Alexa’s broader prose. Alexa’s embedding theorem [2, Theorem 1.1] now applies. $\square$

7. CERTIFICATES AND FORMAL BOUNDARY

The local topology in Lemma 4.3 has a small exhaustive model. An encoding chooses a subset of four triangular faces, whether the tetrahedron is present, a subset of four boundary vertices, and a subset of six boundary edges. This gives

$$16 \cdot 2 \cdot 16 \cdot 64 = 32768$$

raw encodings. These Boolean conditions deliberately form a conservative over-approximation of induced four-vertex pairs arising in triangulated balls.

Lemma 7.1 (Soundness of the finite model). *Every pair (K, K_∂) induced by a four-clique under the hypotheses of Lemma 4.3 is represented by an admissible Boolean encoding.*

Proof. Because the four vertices form a graph clique and $K = T[S]$ is induced, K contains all six edges. Simplicial closure permits the tetrahedron only when all four of its triangular faces are present. An edge of K_∂ has both endpoints in $V(\partial T)$ because ∂T is a subcomplex. Finally, (BT) places every triangle of K on three boundary vertices in K_∂ , and simplicial closure places its three edges there as well. These are precisely the verifier’s admissibility conditions; global manifold restrictions can only remove encodings. $\square$

Exactly 695 encodings are admissible under these local conditions. Exact elimination over $\mathbb{F}_2$ gives the distribution in Table 1.

TABLE 1. Exhaustive check of the local model behind Lemma 4.3.

Quantity	Value
Raw encodings	32,768
Admissible encodings	695
Admissible encodings with $\dim H_2 = 0$	670
Admissible encodings with $\dim H_2 = 1$	25
Maximum dimension	1

Two Python programs obtain these counts through different representations of the same admissibility model. One uses small bit masks; the other constructs explicit simplex sets and reduces boundary-matrix columns. The

Lean check is algorithmically different: it enumerates all sixteen relative 2-chains and counts cycles instead of computing a matrix rank. These computations are regression checks of the hand proof, not premises of it.

Remark 7.2 (Cycle-count criterion). For these four-vertex relative chain complexes, the relative cycle space is an $\mathbb{F}_2$ -vector space containing $\text{im } d_3$. Consequently, $\dim H_2 \leq 1$ is equivalent to the cycle count being at most two when the relative d_3 vanishes and at most four when it is nonzero. The Lean file checks this finite cardinality predicate; it does not formalize the homological equivalence just proved.

The Lean project is pinned to Lean 4.30.0 and has no package dependency beyond the Lean toolchain. It builds without `sorry`, `admit`, or project-declared axioms. Its scope is deliberately limited:

- **FiniteLocal.lean** enumerates the 32,768 encodings and checks the local bound with Lean’s `native_decide`; the resulting finite theorems therefore record the native-decision compiler trust axiom;
- **Cockade.lean** checks an abstract structural-induction schema for recursive safe gluing over explicitly supplied predicates;
- **Reduction.lean** uses that schema: Jørgensen’s decomposition is represented by a base certificate, the manuscript’s no-merging argument is an explicit certificate-propagation input, and HLS is the explicit safe gluing input. Every deep external theorem remains a field of the input structure, whose consistency is witnessed by a trivial model.

Thus **FiniteLocal.lean** is a substantive finite check. The files **Cockade.lean** and **Reduction.lean** are machine-checked dependency diagrams over explicit manuscript-level assumptions. None is a formalization of generic rigidity, PL duality, graph-minor theory, spatial embeddings, or Alexa’s theorem.

The hypotheses hold for every cone over a triangulated 2-sphere. Its skeleton is apex over a planar graph and is therefore K_6 -minor-free and linklessly embeddable. A non-atomic sharp example is obtained by stellarly subdividing a tetrahedron $abcd$ first at an interior point e , and then subdividing the tetrahedron $abce$ at an interior point f . The resulting ball satisfies (BT), and its skeleton is

$$K_6 - df \cong K_5 \oplus_{K_4} K_5, \quad |E| = 14 = 4 \cdot 6 - 10.$$

For the sum clique $S = \{a, b, c, e\}$, deletion leaves the two isolated vertices d, f ; hence the separator bound is attained and the cockade induction is genuinely used. Wood’s bound of at most $4n - 15$ four-cliques in a K_6 -minor-free graph supplies a complementary quantitative restriction on this mesh class [22].

8. SCOPE AND FURTHER QUESTIONS

The theorem removes an apparent minor obstruction only in a sharply defined mesh class. Condition (BT) is used twice: it forces the interior-star cover needed for rigidity, and it controls the relative boundary complex on a four-clique. Dropping it invalidates both bridges. Likewise, the proof does not show that arbitrary MP_1 -cockades are linkless; the separator bound is essential, consistent with the known difference between edge-maximal nIL graphs and edge-maximal K_6 -minor-free graphs [11].

Natural next questions include whether an analogous result holds for suitable homology 3-balls, whether Alexa’s branch-model analysis can be upgraded directly from $K_{3,3,1}$ to K_6 , and which MP_1 -cockades occur as skeletons of tetrahedral balls satisfying (BT). A deeper formalization would also require substantial libraries for simplicial manifolds, generic rigidity, graph minors, and linkless embeddings. In particular, it is open here whether every skeleton in the theorem’s class must be apex. Odeneal and Pavelescu exhibit a non-apex, K_6 -minor-free MP_1 -cockade formed by a K_4 -sum of two apex atoms [13, Future explorations (2)]; deciding whether it, or any analogous non-apex safe cockade, is realizable as such a tetrahedral-ball skeleton would distinguish a geometric obstruction from an artifact of the examples. A recent catalogue of minor-minimal non-apex graphs also proves the order-13 case of Jørgensen’s conjecture, but does not impose this tetrahedral-ball realization condition [14]. Related clique-sum structure for triangles in K_r -minor-free graphs is developed by Albar and Gonçalves [1].

Data and code availability. The preprint, source, exact verifiers, certificates, validation logs, and Lean project are available in the versioned Zenodo record at [doi:10.5281/zenodo.21925574](https://doi.org/10.5281/zenodo.21925574).

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